

Max Flow

Have network flow from s, t and trying to maximize the flow from s to t . Where the arcs have upper bounds.

Max Flow reduces to MCNF

All supplies are 0 at all nodes. Copy the arcs over with the same upper bounds and costs are 0. Then add a new arc from t to s with an infinite upper bound and a cost of -1

Note that this creates a cycle which makes this an equivalent circulation problem.

Dual of Max Flow

$$\begin{aligned} v^* &= \max \sum_{i \in V} b_i \pi_i - \sum_{(i,j) \in U} u_{ij} [-c_{ij}^{\pi}]^+ \\ \text{s.t. } c_{ij}^{\pi} &\geq 0, \quad ((i,j) \notin U) \\ c_{ij} &= 0, \quad (i,j) \in E \\ c_{ts} &= -1 \end{aligned}$$

note since max flow is all supplies are 0 the first term in the objective function is 0

Equivalently:

$$\begin{aligned} f^* &= \min \sum_{(i,j) \in U} u_{ij} [y_i - y_j]^+ \\ \text{s.t. } y_j &\geq y_i, \quad ((i,j) \notin U) \\ y_t &= 0 \\ y_s &\geq 1 \end{aligned}$$

Lemma $f^* \leq c(S, V - S) \quad \forall (S, V - S) \text{ st cut}$

Note these are “cuts” S which is a subset of the nodes V

Lemma Dual of max flow, if $\exists$ finite opt then wlog $\exists$ integer opt

Proof Sketch: Matrix of constraints is transpose of node arc incidence matrix + constraints for upperbounds

Since this is totally unimodular then there exists integer optimal

Lemma $f^* \geq c(S, V - S) \quad \exists(S, V - S)$

Proof If flow is infinite then obvious. If f^* is finite, then y^* (opt sol of dual) and wlog y^* is integer.

$$S = \{i \in V : y_i^* \geq 1\}$$

In general $y^* \in S, y^* \geq 1$ and $y^* \in (V - S), y^* \leq 0$

$$f^* = \sum_{(i,j) \in U} u_{ij} [y_i^* - y_j^*]^+ \geq \sum_{(i,j) \in U \cap \times(S, V-S)} u_{ij} [y_i^* - y_j^*]^+ \geq \sum_{(i,j) \in \times(S, V-S)} u_{ij} = c(S, V - S)$$

In other words we only look at the arcs that cross the cut and have an upper bound.

We know that all arcs that cross the cut must have an upperbound because if they did not then the cut would be invalid? (need to check textbook)

Thm Max Flow Min Cut theorem

Value of max $s - t$ flow = min capacity of $s - t$ cut.

Vertex Cover Reminder

$$V' \subset V : V' \cap e \neq \emptyset \forall e$$

In other words every edge has at least one vertex in V'

Matching set $M \subset E$ s.t.

$e, f \in M \quad e \cap f = \emptyset$ which means that a subset of edges where they have no vertices in common

Note that the smallest matching is just an empty set. More meaningfully however is to find $\max |M|$

Def Bipartite Graph $G = (V_1 \cup V_2, E) \wedge (V_1 \cap V_2 = \emptyset) \wedge \forall e (e \cap V_1, e \cap V_2 \neq \emptyset)$ **Proof** in bipartite graphs, $\min |V'| = \max |M|$

Pf Convert the bipartite graph into a directed graph with the same edges from V_1 to V_2 . Add a source node s to V_1 with upperbounds of 1 and similarly from V_2 to target node t with upper bounds of 1. All original edges have infinite bounds.

$$1. f^* = \max |M|$$

$$i. f^* \leq |M^*|$$

f^* is finite (see cut $(\{s\}, V - \{s\})$ has finite capacity)

flow of paths (from s to t) + cycles (ϕ because directed acyclic graph).
Any path must pass through the original arcs and these selected arcs are
a matching (otherwise upperbound constraints will be violated). Thus
given f^* gives a matching so $f^* \leq |M^*|$

2. $\min c^*(S, V - S) = \min |V'|$

2a. given a Vertex Cover V' then $\exists$ s-t cut $(S, V - S)$ s.t. $|V'| = c(S, V - S)$

A cut will be on the S side include the source and any vertices not in V' but in V_1
and the vertices in V' and in V_2 . All others are then consequently in $V - S$.

The capacity of this cut is $|V'|$ (because the arcs from s to $V_1 \cap V'$ and $V' \cap V_2$)
plus the arcs that go from V_1 to V_2 but are not in V' (which is actually 0).

By contradiction, $\exists$ original arc $(i, j) \in X(S, V - S)$ thus $i \in V_1, j \in V_2$. Since it
is an original arc and we constructed our cut in a specific way, then this is saying
that $i \notin V' \wedge j \notin V'$ which is a contradiction from the definition of V' . Thus the
capacity of the cut is $= |V'|$

2b. given s-t cut $(S, V - S)$ of finite capacity then $\exists$ vertex cover $|V'|$ s.t. $|V'| = c(S, V - S)$

From our cut we perform the same methods for choosing our VC nodes (not in cut
on V_1 side and in cut on V_2 side). $V' = (V_1 - S) \cup (V_2 \cap S)$

Since cut is finite capacity then there is no original arcs that cross over our cut
because we assigned the original arcs to have no upper bound.

Thus by max flow min cut, that $|f^*| = \min c(S, V - S)$ so $\max |M'| = \min |V'|$